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ORIGINAL ARTICLE

Short-term effects of normobaric hypoxia on the human spleen

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Abstract Spleen contraction resulting in an increase in circulating erythrocytes has been shown to occur during apnea. This effect, however, has not previously been studied during normobaric hypoxia whilst breathing. After 20 min of horizontal rest and normoxic breathing, five subjects underwent 20-min of normobaric hypoxic breathing (12.8% oxygen) followed by 10 min of normoxic breathing. Ultrasound measurements of spleen volume and samples for venous hemoglobin concentration (Hb) and hematocrit (Hct) were taken simultaneously at short intervals from 20 min before until 10 min after the hypoxic period. Heart rate, arterial oxygen saturation (SaO_2) and respiration rate were recorded continuously. During hypoxia, a reduction in SaO_2 by 34% ($P < 0.01$) was accompanied by an 18% reduction in spleen volume and a 2.1% increase in both Hb and Hct ($P < 0.05$). Heart rate increased 28% above baseline ($P < 0.05$). Within 3 min after hypoxia SaO_2 had returned to pre-hypoxic levels, and spleen volume, Hb and Hct had all returned to pre-hypoxic levels within 10 min. Respiratory rate remained stable throughout the protocol. This study of short-term exposure to eupneic normobaric hypoxia suggests that hypoxia plays a key role in triggering spleen contraction and subsequent release of stored erythrocytes in humans. This response could be beneficial during early altitude acclimatization.

Keywords Hypoxia · Spleen · Hemoglobin · Hematocrit · Ultrasound

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Introduction

The spleen is known to serve as a dynamic reservoir for red blood cells in many species (Barcroft and Stephens 1927; Kramer and Luft 1951; Hurford et al. 1996). In humans, the splenic reservoir is relatively small compared to body size but still augments the body's circulating hematocrit (Hct) and hemoglobin (Hb) concentration when the spleen contracts. These spleen-associated increases have been shown to occur during apnea (Schagatay et al. 2001; Bakovic et al. 2005) and are reversible within minutes (Schagatay et al. 2005). Similar effects occur during high-intensity exercise (Laub et al. 1993). The benefit of such a transient increase is an improved oxygen carrying capacity during situations where it would be particularly helpful, while avoiding polycythemia when it is not necessary through expedient storage of extra red blood cells by the spleen.

One such situation where fast Hb elevation would be helpful is acute hypoxia. Rats have shown an increase in Hb originating from the spleen in response to acute intermittent hypoxia, which is also reversible and mediated by alpha2-adrenoreceptors (Kuwahira et al. 1999). Kramer and Luft (1951) also showed in a study on dogs that Hb increased directly correlated to the size of the spleen following severe hypoxia. In humans, Richardson et al. (2005a) have suggested hypoxia to be involved in initiating the spleen-associated Hb and Hct increase during breath holding, together with one or more other stimuli.

Human splenic contraction is not well-investigated in general hypoxic situations, and previous work has focused on chronic exposures to altitude. Using ultrasonography, Sonmez et al. (2007) confirmed that 3–6 months exposure to altitude through relocation of lowlanders resulted in reduced splenic volume and concomitant increases in Hb and Hct, but no specific focus was directed toward the ascent and descent phases. Richardson and associates (2007) discovered that the transient elevation in Hb during apnea was attenuated and eventually disappeared as individuals travelled to progressively higher altitudes, during which time an overall polycythemia occurred. During descent, the Hb increase during apnea was greater than at similar altitudes during ascent. This suggests that the spleen may contract tonically with increasing altitude, eventually reaching a "fully contracted" state which would not produce any further Hb elevation even during additional hypoxic insult i.e. apnea. During descent, after three weeks of increasing altitude, splenic reuptake of unnecessary red blood cells may have resulted in a greater Hb increase through expulsion of its contents during apnea.

There has been no previous study of the short-term effects of a hypoxic stimulus during regular breathing on the spleen. The aim of this study was therefore to reveal whether spleen contraction and related increases in Hb and Hct occur during eupneic exposure to short-term normobaric hypoxia.

Methods

Subjects

Five volunteer subjects (4 men, 1 woman), of mean ± SD age 27 ± 3 years, height 1.85 ± 0.1 m, and mass 89 ± 21 kg participated in the study. All were healthy non-smokers who trained 5 ± 3 h per week for physical fitness, and had a vital capacity of 6.1 ± 0.7 L. None had been at high altitude for at least 3 months prior to the study. The study protocol was approved by the ethics committee at Umeå University, Sweden and complied with the 2004 Declaration of Helsinki for research involving human subjects.

Experimental protocol

Testing took place at Härnösand regional hospital. After 20 min of horizontal rest and normoxic breathing at sea level, subjects underwent 20-minutes of hypoxic breathing (hypoxia) followed by 10 min of normoxic breathing. The hypoxia was induced via mask (Hypoxico, Hypoxico Inc., New York, USA) to an oxygen content of 12.8%, which simulated approximately 4,100 m above sea level. All subjects remained prone and resting during the entire protocol. For analysis of Hb and Hct 1.5 mL blood samples were taken via intravenous catheter (Venflon Pro, Beckton-Dickson AB, Helsingborg, Sweden) placed at the antecubital region, at the time periods illustrated in Fig. 1. Ultrasound measurements of the spleen (via HDI 3000, ATL A, Philips Company, Bothwell, WA, USA) were taken in three axes by a trained professional, simultaneously with the blood samples. Arterial oxygen saturation (SaO_2) and heart rate (HR) were measured continuously throughout the protocol via pulse-oxymeter (WristOx 3100, Nonin Medical Inc., Plymouth, MN, USA). Breathing movements were also recorded through the protocol using a lab-developed pneumatic chest bellows.

The total volume of blood (including waste volume) obtained from each subject was approximately 40 mL, and the injected saline was approximately 20 mL. The analysis of Hb and Hct was conducted via automated blood analysis unit (Micros 60 Analyzer, ABX Diagnostics, Montpellier, France).

Data analysis

Spleen volume was calculated using a formula developed by S. Pilström (personal communication): $L\pi(WT - T^2)/3$, where (L) is spleen length, (W) width and (T) thickness obtained from the ultrasound measurements. The formula is based on the average shape of the spleen observed on the ultrasonic images and describes the difference between two ellipsoids divided by 2. Control values for spleen volume, Hb and Hct were established immediately prior to the onset

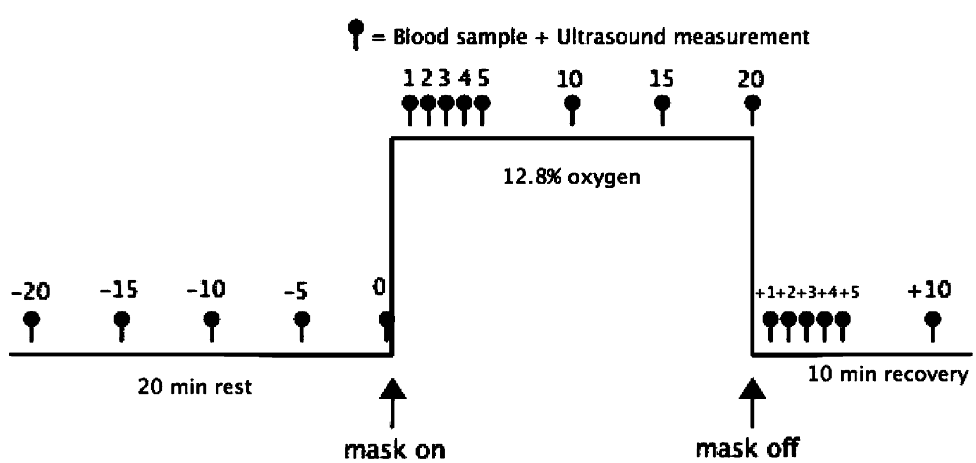

Fig. 1 The time course of the test protocol, illustrating the hypoxic and normoxic periods and sampling points for blood and ultrasound measurements

of hypoxia, and then compared with values from hypoxia and post-hypoxia. For SaO_2 and HR, the control value was established as a 60 s average recorded 90–30 s before the onset of hypoxia, and compared with 30 s average values taken at the same time points as the ultrasound and blood measurements. All values were log-transformed before comparison via *t* test. 90% confidence intervals and *P* values were calculated for all comparisons. Pearson *r* correlations were performed between spleen volume and blood parameters, and both *P* values and the likelihoods for true values based on Cohen's threshold (Cohen 1988) were calculated via purpose-made spreadsheet (Hopkins 2002).

Individual observed reductions in spleen volume during the hypoxic period were used to represent expelled blood volumes, with a Hct of 80% (Laub et al. 1993). The effects of adding this to the circulating blood was calculated based on the measured individual Hct before hypoxia and an estimated circulating blood volume based on height and weight (Nadler et al. 1962). The mean estimated Hct increase for all measurements during the hypoxic period was then compared to the measured increase in Hct to calculate the contribution from the spleen.

Results

Results are reported as mean ± SD for anthropometric data, and as mean (90% confidence limits) for changes and comparisons, with *P* values.

Pre-hypoxia

Control values were established for SaO_2 at 96.6 ± 2.0%, HR at 62 ± 17 bpm, and respiration rate at 9.5 breaths per minute. During the pre-hypoxic 20 min period of horizontal rest, spleen volume increased and Hb and Hct decreased as the subject's blood volume and capillary exchange equilibrated as a result of the change in position. Prior to the onset of hypoxia, spleen volume was 405 ± 133 mL, Hb was 134.4 ± 9.0 g/L and Hct was 39.8 ± 3.2% BV.

Hypoxia

SaO_2 fell and HR increased promptly during the initial 5 min of hypoxic exposure, to reach 33.9% (17.2) below baseline for SaO_2 (64.4%; $P < 0.01$) and 28.1 (19.5)% increase above baseline for heart rate by 20 min ($P < 0.05$). Respiration rate was 10.3 breaths per minute by the end of the hypoxic exposure (NS).

Mean spleen volume was reduced by 13.3% (6.4) after 3 min of hypoxic breathing ($P < 0.01$; Fig. 2). At 15 min it had been reduced by 17.9% (8.7) and after 20 min the volume was reduced by 17.6% (8.7) below the control value (both $P < 0.01$).

Hb had increased by 2.1% (1.0) to 137.2 ± 9.0 g/L at 3 min ($P < 0.05$), and was 136.2 ± 9.9 g/L after 20 min of hypoxic breathing (Fig. 2). Hematocrit followed the same response pattern as Hb, increasing by 2.1% (0.9; $P < 0.01$)

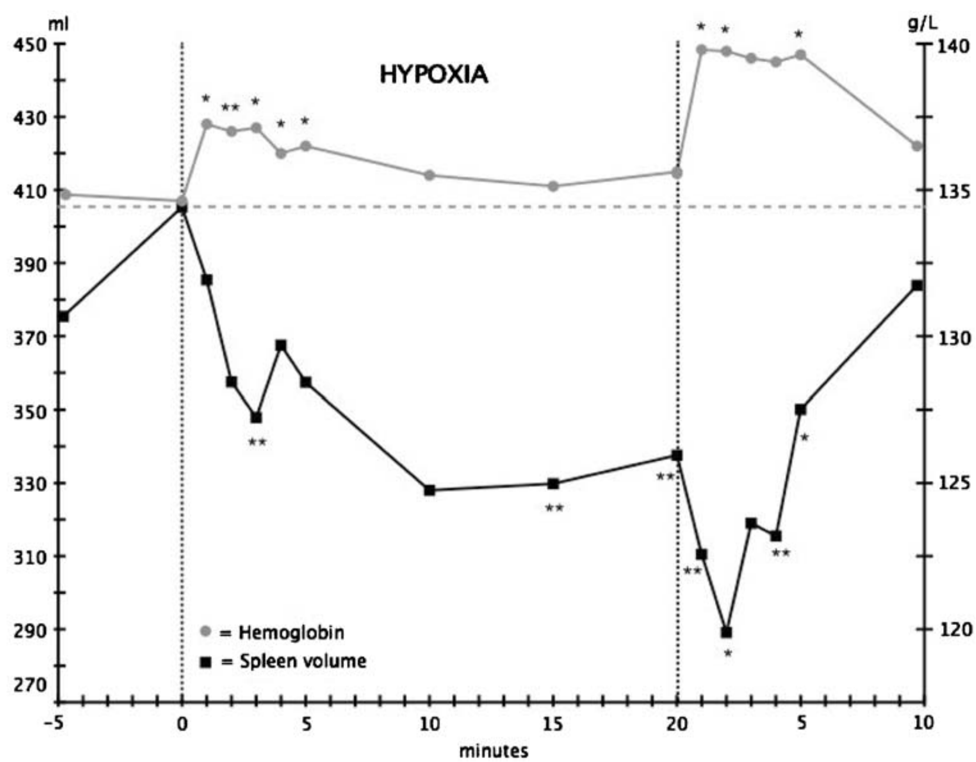

Fig. 2 Spleen volume and Hb during pre-hypoxia, hypoxia and post-hypoxia for $N = 5$. For comparison to control values, $P < 0.05$ is indicated by * and $P < 0.01$ by **

to 40.6 ± 3.4% BV at 3 min hypoxia and was 40.2 ± 3.3% BV after 20 min of hypoxic breathing.

Post-hypoxia

There was a prompt return of SaO_2 and HR to the control values following the onset of the post-hypoxic period, and HR values were −3.8% (10.6; NS) and −12.7% (11.5; NS) compared to control at 5 and 10 min post-hypoxia, respectively. Respiration rate was 9.2 breaths per minute, 10 min after exposure.

Spleen volume was still reduced by 22.2% (11.3; $P < 0.01$) and 13.6% (11.0, $P < 0.05$) at 1 and 5 min post-hypoxia, respectively (Fig. 2). Corresponding SaO_2 values were −6.1% (3.6; $P = 0.05$) and −2.9% (4.5; NS) compared to the control value. At 10 min post-hypoxia spleen volume was −5.8% (9.7; NS) and SaO_2 was −1.4% (1.8; NS), compared to control values.

Compared to the control value, Hb was still elevated by 3.4% (2.0; $P < 0.05$) and 3.3% (2.0; $P < 0.05$) at 1 and 5 min post-hypoxia, respectively, but gradually decreased throughout the post-hypoxic period and by 10 min post-hypoxia was 1.8% (2.2; NS) above control value (Fig. 2). Hct followed a similar pattern, with an increase of 3.2% (2.1; $P < 0.05$) and 2.6% (±2.1; NS) at 1 and 5 min post-hypoxia, respectively, gradually decreasing to 0.9% (1.7; NS) from the control value at 10 min post-hypoxia.

Correlation between spleen volume and blood parameters

The Pearson r correlation between spleen volume and Hb was −0.51 ($P < 0.05$), representing 74% likelihood that the true value of the correlation is clinically negative according to Cohen's threshold of 0.1. For the correlation between spleen volume and Hct of −0.42 ($P < 0.1$), this represents 69% likelihood that the true value of the correlation is clinically negative according to Cohen's threshold of 0.1. The spleen was estimated to contribute to 60% of the observed overall increase in Hct during hypoxic exposure.

Discussion

This study shows that spleen contraction develops rapidly upon onset of normobaric hypoxia in humans. An ejection of stored erythrocytes results, as demonstrated by the corresponding elevations in Hct and Hb. These responses are reversed within approximately 10 min upon return to normoxia. This result suggests, together with other recent observations (Richardson et al. 2007) that spleen contraction may be part of the early acclimation to altitude in humans, whereby increases in circulating hematocrit could help overcome the initial phase of hypoxia before erythropoietic effects may develop. These early elevations in Hct and Hb were previously suggested to be associated with changes in plasma volume caused by the hypoxia-induced responses of sodium- and water-regulating hormones (Schmidt 2002) or dehydration due to increased ventilation (Heinicke et al. 2003). However, in the current study it appears that the spleen is an important source of extra circulating erythrocytes during hypoxia.

The spleen contraction generally follows the development of arterial hypoxia suggesting, in accordance with earlier findings, that hypoxia may be an important input for its development (Richardson et al. 2005a). In that study, removal of the apneic hypoxia by inspiration of oxygen attenuated the rise in Hb and Hct, but some response remained showing that other factors are also involved. A possible contributing factor in the apneic situation could be hypercapnia, but in the current context hypercapnia is highly unlikely, as the respiration rate was unchanged during the hypoxic exposure. Catecholamines or sympathetic nervous system activity related to stress might also contribute to splenic contraction, although the conduct of the current experiment was not considered by subjects to be stressful (personal communication). Apneic situations are likely perceived as more stressful, but changes in Hb and Hct in some previous apneic studies (Schagatay et al. 2005; Richardson et al. 2005b) were of a similar magnitude as in the present study.

The decrease in SaO_2 was most intense during the initial 5 min, which may explain the powerful spleen responses during this period. Interestingly, it appears that both the initiation and interruption of the hypoxic exposure initially evoke a spleen contraction, as indicated both by the spleen volume and blood boosting response. We speculate that this could either be a response to the change in inspired oxygen, irrespective of direction, or a general arousal response associated with the placement and removal of the breathing mask. However, the direction of the response after the initial phase in these two situations is clearly different, showing that during hypoxia spleen contraction prevails.

The question if there are any physiological differences between hypobaric hypoxia compared to normobaric hypoxia for the same ambient PO_2 is still under debate. In addition to lowered SaO_2, Savourey and associates (2003) reported changes in mainly respiratory parameters at 4,500 m altitude, such as increased breathing frequency, tidal volume and minute ventilation, resulting in hypocapnia and a higher blood alkalosis. While blood chemistry and tidal volume were not measured in the current protocol, respiration rate remained unchanged during the hypoxia, suggesting that no hypercapnia occurred.

This laboratory study supports the observations of a previous study in three climbers (Richardson et al. 2007) where apnea-induced elevations in Hb were found to diminish with increased altitude, conceivably due to emptying of the spleen with increasing hypoxia. The present study suggests that even during normobaric eupnea, hypoxia is a trigger for spleen contraction and increased volume of circulating erythrocytes. This may provide a rapid means of increasing oxygen carrying capacity during early phases of altitude adaptation as well as during acute hypoxic situations.

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